Annex B1: Characterization of the 50 Kg Load Cell.

1. Introduction

This annex outlines the methodology employed to characterize the 50 Kg load cell integrated into the machine. The analysis includes the determination of the relationship between the applied force and the output signal, as well as the evaluation of linearity on the working range. The results ensure the accuracy of measurements obtained during experimental testing.

2. Materials and Methods

2.1. Equipment Used

- Load cell: Model DYLY106, with a maximum nominal capacity of 50 kg with a nominal output of 2.0mV/V, and an accuracy of +/- 0.1%.
- Load Cell Amplifier Transmitter: A WXD-D70 (JY-S60) model was selected, optimized for this kind of load cell. The device accepts an input signal with sensor sensitivity 2.0 mV/V up to 20mV and provides a DC output of 0–5V / 0–10 V / 4–20 mA
- Power supply: 24 V.
- Data acquisition system: Arduino-based interface with custom characterization software.
- A vertical press of initial compression load of 4.04 Kg.
- A hook of 0.76 Kg.
- Calibration standards: A set of precision weights of 1, 2, 4, and 10 Kg, previously verified using a digital calibrated scale (OHAUS Ranger 3000).

Fig. 1. Hardware for load cell characterization.

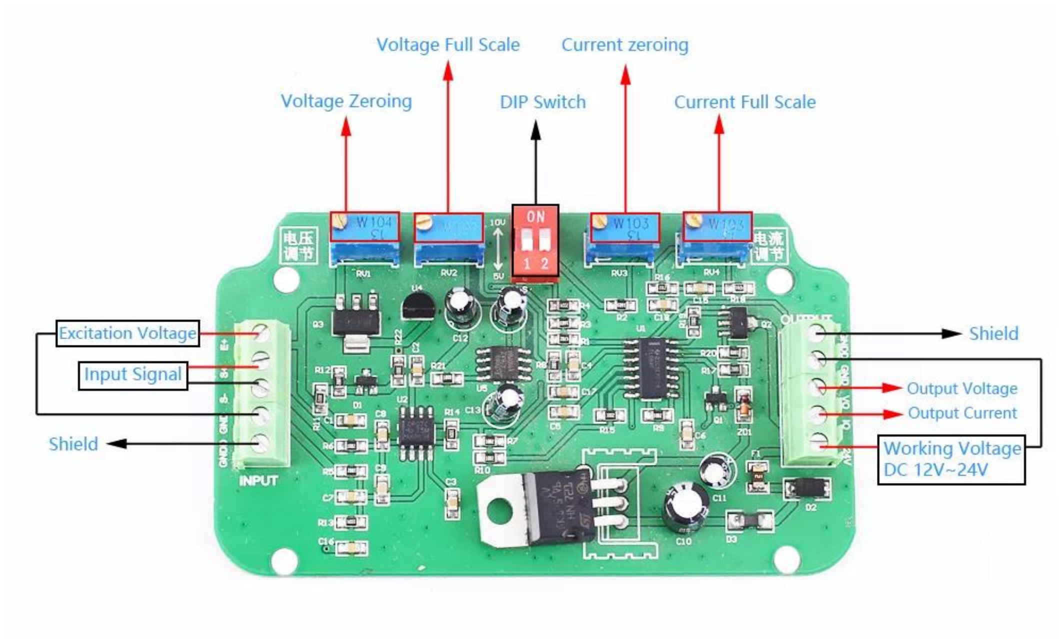

Fig. 2. Load cell amplifier transmitter wiring. [1]

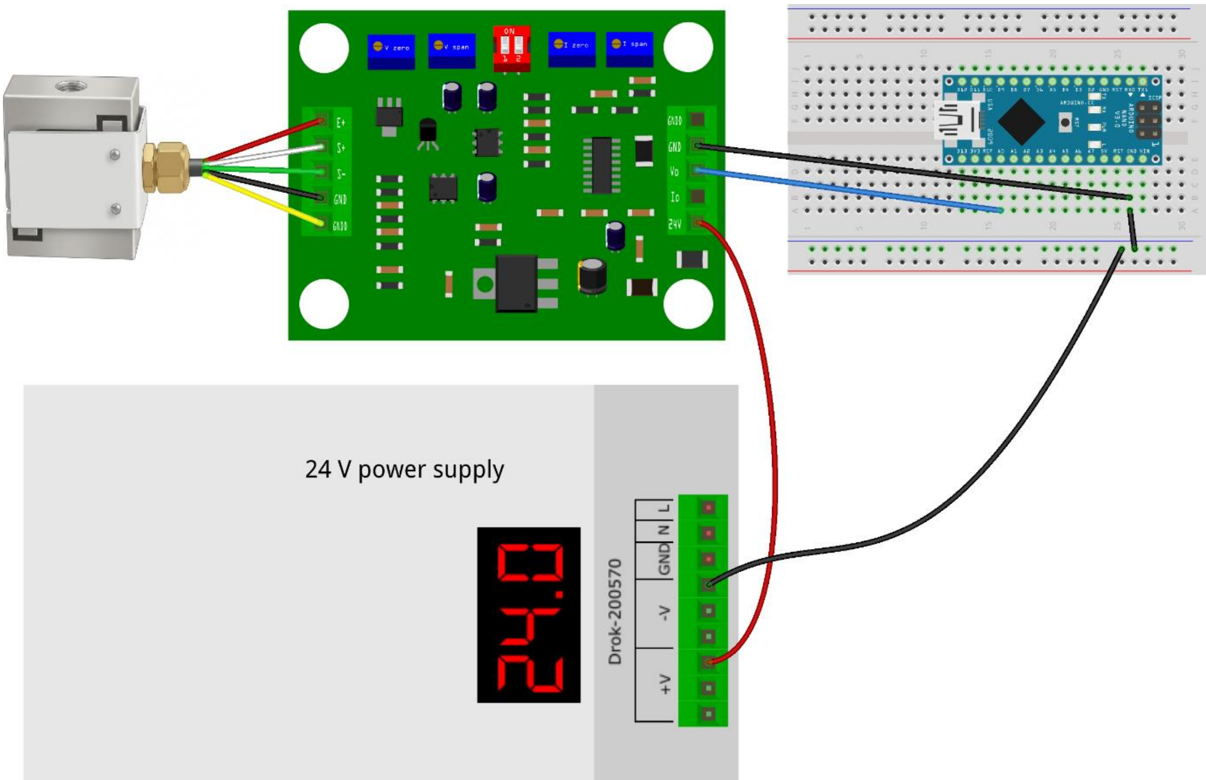

Fig. 3. Breadboard layout for load cell characterization.

2.2. Experimental Procedure

Following the wiring instruction of all the components, the load cell was wired to the amplifier and the Arduino Nano board, who was connected to a PC, for data acquisition (See Fig. 1 to 3).
A vertical traction to compression system was implemented, incorporating a suspension mechanism and the press. Then Incremental loads were applied using the calibration weights, ascending to the load cell’s 80% maximum capacity. For each measurement, the output voltage was recorded (See Fig. 4).

The span voltage calculation, as provided by ATO in the Load Cell Transmitter User Manual, was applied [2]:

*Span voltage = Object weight / Load Cell capacity * 5V*

This formula was integrated into the Arduino-based data analysis software to generate the specific characterization detailed herein. Finally, the acquired data were fitted using linear regression.

Fig. 4. Gradual increment of calibration weights.

3. Results

The characterization provided the relationship between the output voltage and vs the weight load applied, also confirms that the load cell meets the precision and repeatability requirements necessary for the project.

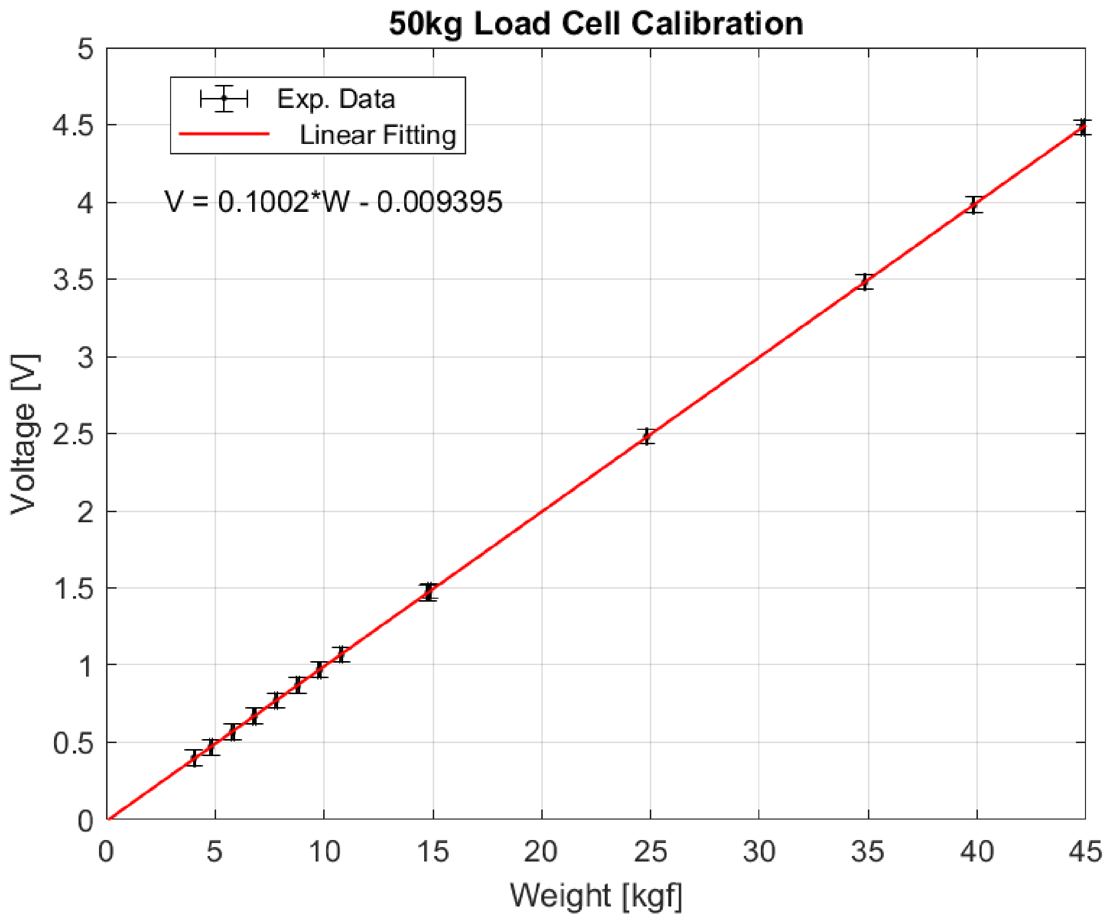

Fig. 5. Linear fit for the output voltage vs the weight.

As it can be seen in the Fig. 5, the ratio of Output voltage/Load is 0.1002 V/kgf or 0.1002 V/kg +/- 1%. Doing the conversion to N/V (adjusted for Bogotá's gravitational acceleration of 9.774 m/s^2), it becomes 97.54 N/V +/- 3%, which it's the final ratio that going to be set into the Arduino-based data acquisition software. The R-square is 1 in the evaluated range, so it proves the linearity of the load cell.

Either for the characterization or the experiments, it is recommended not to exceed the 90% of the maximum load capacity to avoid losses on the measurement accuracy or linearity or even mess up the load cell.

Annex B2: Characterization of the 1 Kg Load Cell.

1. Introduction

The previous section presented the specifications for characterizing the 50 kg load cell. The methodology followed here is broadly analogous to that described therein, with a few changes described below.

2. Materials and Methods

2.1. Equipment Used

- Load cell: S-type Miniature Model DYLY-106, with a maximum capacity of 1 kg with a nominal output of 1.0mV/V, and an accuracy of +/- 1%
- Load cell signal amplifier: An Shitailinge voltage signal transmitter model was selected, optimized for this application due to its high performance. The device accepts an input signal of 0.8–3 mV and provides a DC output of 0–5V.
- Power supply: 24 V.
- Data acquisition system: Arduino-based interface with custom characterization software.
- Calibration standards: A set of precision weights (5g, 10g, 20g, 50g, 99g, 100g, 998g), previously verified using a digital scale (OHAUS Scout Pro).

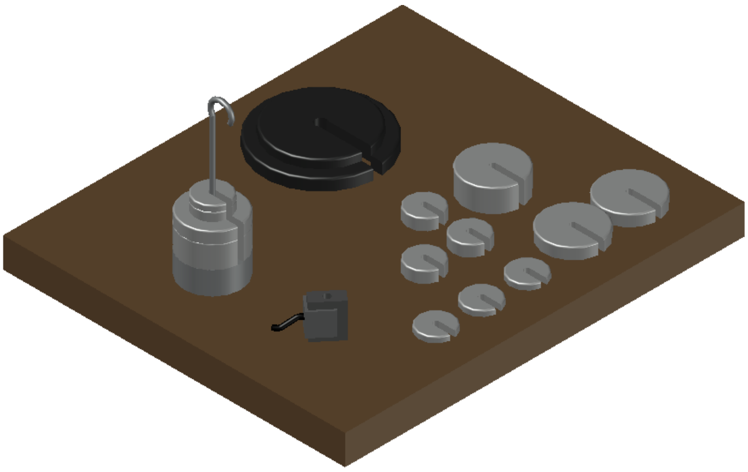

Fig. 6. 1kg load cell characterization assembly.

2.2. Experimental Procedure

This time, the load cell was characterized under direct compressive loading conditions, where calibration weights were progressively added in incremental steps, in contrast to the tensile-compression loading methodology described in Appendix B1 (See Fig. 6 and Fig. 7). The load cell was wired to the signal amplifier and subsequently to the Arduino Nano board, which was connected to a PC for data acquisition. As in the previous characterization, wires and connections were made following the wiring instruction of all the components, shown in Fig. 8 and Fig. 9.

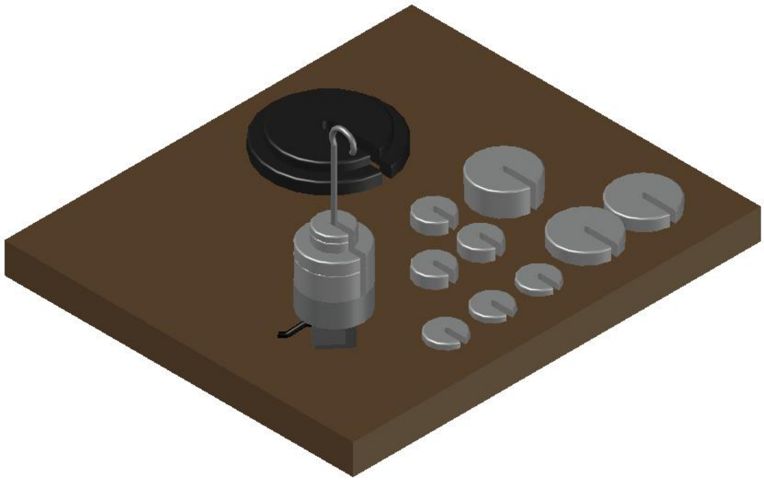

Fig. 7. Gradual increment of calibration weights on the 1kg Load Cell

The span voltage calculation, as provided by ATO in the Load Cell Transmitter User Manual, was applied [2]:

*Span voltage = Object weight / Load Cell capacity * 5V*

The same Arduino software framework from the initial calibration was maintained, apart from the modified span voltage calculation mentioned above. As with the previous tests, linear regression analysis was applied.

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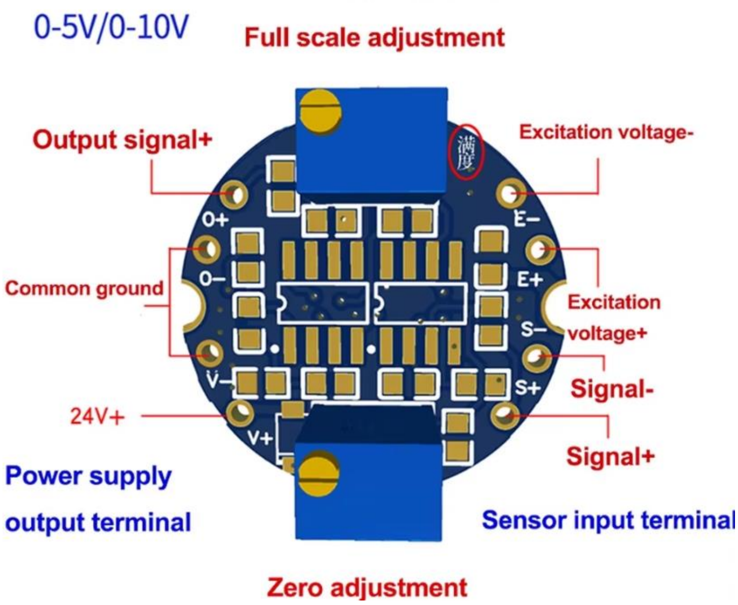

Fig. 8. Load cell signal amplifier wiring. [3]

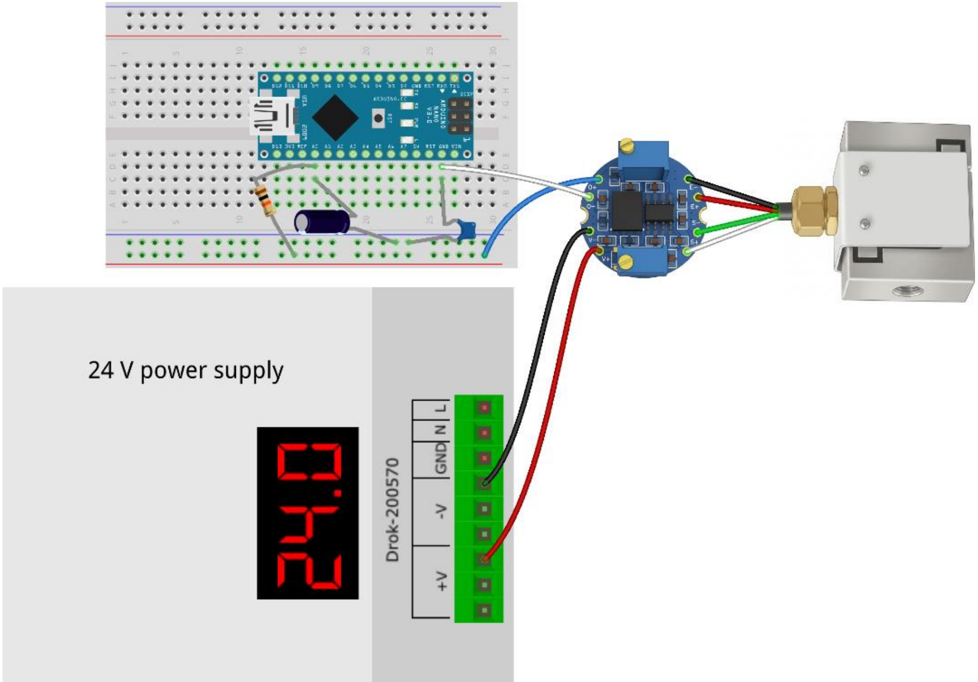

Fig. 9. Breadboard layout for 1 kg load cell characterization.

3. Results

Like in the 50kg load cell's characterization, the analysis provides the relationship between the output voltage and vs the weight load applied and confirms that the 1kg load cell meets the precision and repeatability requirements necessary for the project.

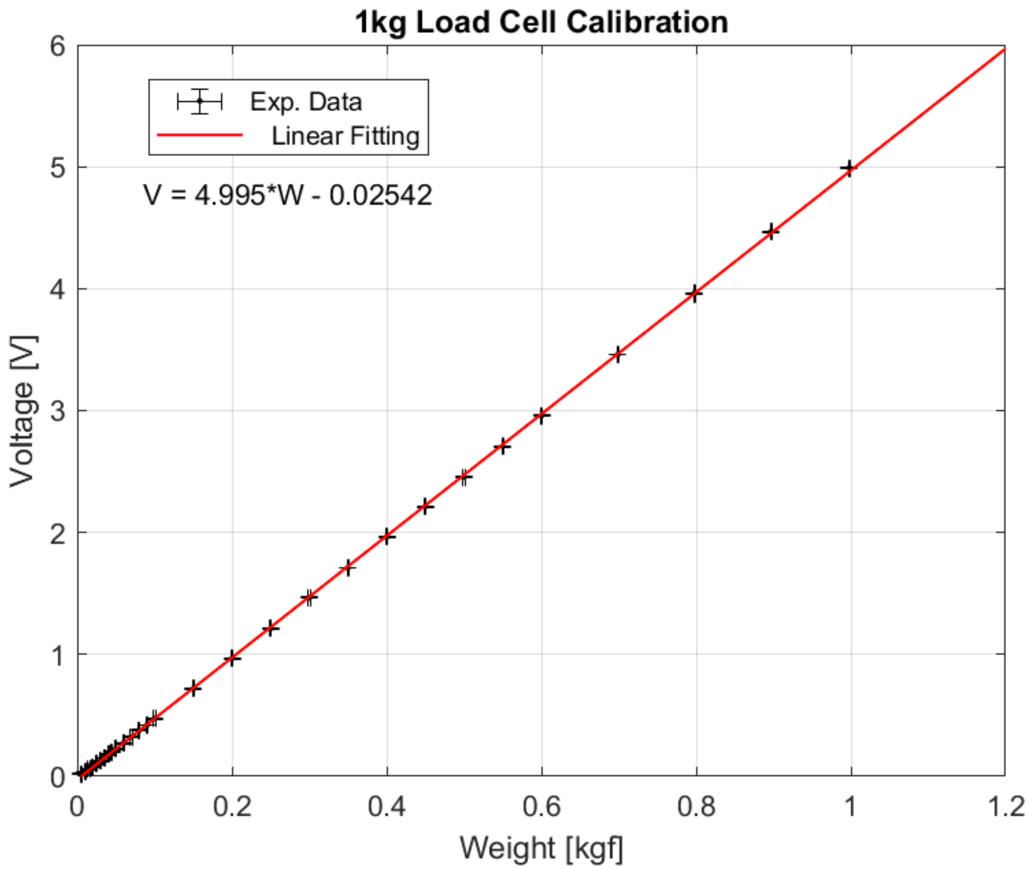

Fig. 10. Linear fit for the output voltage vs the weight.

As it can be seen in the Fig. 10., the ratio of Output voltage/Load is 4.995 V/kgf or 4.995 V/kg +/- 2%. Doing the conversion to N/V (adjusted for Bogotá's gravitational acceleration of 9.774 m/s^2), it becomes 1.96 N/V +/- 5%, which it's the final ratio that going to be set into the Arduino-based data acquisition software. The R-square is 0.9999 in the evaluated range, so it proves the linearity of the load cell.

For both the characterization and the experiments, it is recommended not to exceed 95% of the maximum load capacity, to avoid losses in measurement accuracy or linearity, or even damage to the load cell. It's critically important to not exceed the input power supply, since the signal amplifier could easily burn out.

References:

[1] “Load Cell Transmitter JY-S60,” Micro Robotics, [Online]. Available: https://www.robotics.org.za/JY-S60-12V. [Accessed 31 July 2026].

[2] ATO international, “Load Cell Transmitter User Manual,” [Online]. Available: Load-cell-transmitter-user-manual-ATO-LCTR-OA (https://www.ato.com/Content/doc/Load-cell-transmitter-user-manual-ATO-LCTR-OA.pdf?srsltid=AfmBOopccpYjDacgM1EVRntKrzYOnJrbMtuJf6hWH6vLKNim6oE23lzz). [Accessed 31 July 2026].

[3] STAILNGE-Shitailinge, “Load Cell Amplifier Signal Transmitter Module Current 4-20mA 0-10V Output RS485 Modbus RTU Tension Compression Sensor,” AliExpress Alibaba Group, [Online]. Available: Load Cell Amplifier Signal Transmitter Module Current 4-20mA 0-10V Output RS485 Modbus RTU Tension Compression Sensor - AliExpress 502 (https://n9.cl/h2mejt). [Accessed 31 July 2026].